

Mohammad Ismail Nizami

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Professional Summary

AI/ML undergraduate with hands-on experience building and deploying end-to-end machine learning applications using Python, Scikit-learn, TensorFlow, XGBoost, SHAP, and Streamlit. Skilled in exploratory data analysis, data preprocessing, feature engineering, model comparison, cross-validation, hyperparameter tuning, evaluation, and explainable AI. Additionally trained in Generative AI, Retrieval-Augmented Generation (RAG), and NLP. Seeking internship opportunities in Machine Learning, Data Science, Data Analytics, or Python development.

Education

Shri Shankaracharya Technical Campus, Bhilai
B.Tech in Computer Science and Engineering (AI & ML)

Bhilai, Chhattisgarh
2023 – 2027

Projects

Student Placement Prediction [GitHub](#) | [Live Demo](#)

- Developed an end-to-end machine learning application for predicting student placements using academic, technical, and skill-based features.
- Compared Logistic Regression, KNN, Decision Tree, and Random Forest using feature engineering, cross-validation, and model evaluation.
- Deployed the optimized Logistic Regression model on Streamlit with placement prediction and personalized recommendations.

Dog Breed Classification [GitHub](#) | [Live Demo](#)

- Developed a 120-class dog breed classifier using MobileNetV2 transfer learning and fine-tuning, achieving 78.88% Top-1 and 97.11% Top-5 accuracy.
- Developed an image prediction pipeline with preprocessing, confidence scoring, and Top-5 predictions.
- Deployed the trained TensorFlow/Keras model as an interactive Streamlit web application with real-time image upload and prediction.

Air Quality Prediction [GitHub](#) | [Live Demo](#)

- Developed an end-to-end Air Quality Index (AQI) prediction system using data preprocessing, feature engineering, and pollutant analysis.
- Compared Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and a Hybrid Ensemble model using cross-validation and performance evaluation.
- Deployed the Hybrid Ensemble model on Streamlit with SHAP-based explainability, interactive visualizations, and real-time AQI prediction.

Experience

SAIL Bhilai Steel Plant

Vocational Trainee, Human Resources Development Department

Aug 2025 – Sep 2025
Bhilai, India

- Worked on a Retrieval-Augmented Generation (RAG) chatbot during four-week industry training.
- Developed an XML-to-JSP converter using Python for automated code transformation.
- Received an Excellent performance evaluation for project work and overall performance.

KodeZen Technologies

Vocational Trainee – Data Analytics

Jun 2026 – Jul 2026

- Completed a four-week vocational training program focused on Data Analytics.
- Strengthened practical understanding of data collection, cleaning, analysis, and interpretation workflows.
- Applied data analytics concepts through guided exercises and practical training activities.

Technical Skills

Programming: Python, C/C++ (Basic), Java (Basic)

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Streamlit, SHAP

Machine Learning: Data Preprocessing, EDA, Feature Engineering, Classification, Regression, Cross-Validation

Deep Learning: Neural Networks, Transfer Learning, Image Classification

Generative AI: GenAI fundamentals, Retrieval-Augmented Generation (RAG), NLP basics

Tools: Git, GitHub, Jupyter Notebook, Google Colab, GridSearchCV, RandomizedSearchCV

Research & Presentation

- Presented the research paper “Improving Air Quality Prediction Using Hybrid Ensemble Learning with SHAP-Based Explainability” at E-Summit 2026.
- Proposed a hybrid ensemble learning approach and applied SHAP explainability to interpret individual AQI predictions.
- Received a Certificate of Participation for presenting the research work at the multidisciplinary student conference.

Certifications

- Data Science & AI Masters 2026: From Python to GenAI (116 Hours) – Udemy — covers Generative AI, RAG, and NLP
- Complete A.I. & Machine Learning, Data Science Bootcamp (44 Hours) – Udemy
- Vocational Training Certificate – SAIL Bhilai Steel Plant
- Data Analytics Vocational Training (4 Weeks) – KodeZen Technologies